

ARC / CIRCLE 7-WEEK HACKATHON · AGENTIC ECONOMY TRACK · FINAL SUBMISSION

ACR – the Arc Compute Rate

Machine commerce just got its **SOFR** – and it prints its own **attack cost**.

A manipulation-resistant reference rate for machine services, live on Arc testnet – and the cash-settled futures venue that trades on it.

arc-compute-rate.vercel.app · github.com/kaustubh76/ACR---Arc-Compute-Rate

THE PROBLEM

Agents trade machine services with **no trustworthy price**

- Agents already pay per call for inference, GPU time, and data – but there is **no benchmark** to settle contracts against or hedge with.
- The raw payment exhaust is **noisy, batched, and adversarial**.
- A naive volume-weighted average is an open invitation: our own red team moves it **+122%** with wash trades.

No benchmark means no hedging, no term markets, no credit. Machine commerce is stuck at spot.

Meet the hedger: an agent business with a floating USDC burn

- **Portia**, an autonomous paralegal fleet, resells LLM work at fixed prices but pays for inference per call – **~100M tokens/mo × 0.49236 \$/1k tokens** (the ACR-INF print) **≈ \$49,236/mo**, all floating. One +10% month is **+\$4,924** of unbudgeted burn.
- The fix: go **long ACR-INF futures**. Cash-settled – at expiry the venue **freezes the freshest oracle print** (≤ 2 h old, enforced on-chain) and pays the difference in USDC. **No delivery, no GPU repossession, no seller cooperation.**
- A hedge is only as good as its settlement print: every ACR print ships its **CI** and its **attack-cost-per-bp**, so both sides can read – on-chain – what bending the settle would cost.
- **This loop is live on Arc testnet:** `ACRFutures 0x29d97c62...82642fe` – **three books** (ACR-INF, ACR-GPU, ACR-DATA), 10× multiplier, 2000 bp initial margin on this deployment, socialized-loss clearing, settling against the same ACROracle.
- **And an agent already runs Portia's playbook.** It pays \$0.0001 for the print over x402, reads its own book, and trades the gap to its mandate – from a Circle agent wallet, unattended. **It**

Every Circle wallet model, doing a job – not ticking a box

AGENT	WALLET	PAYS FOR	DECIDES
press	Developer-Controlled	gas to sign 3 prints/hr	nothing, deliberately
maker · taker	Developer-Controlled ×2	collateral, roll gas, hourly fill	size, against both sides' margin
hedger	Agent wallet (SCA trades, EOA pays)	the print, then its own margin	everything – and refuses rather than clamps
reader	User-controlled SCA, PIN	nothing: Gas Station sponsors	their own trade, their own exit
CLI buyer	raw EOA + x402-batching	\$0.0001 × 13 resources	which listings to buy

We consume Circle's Skills and publish one back (**skills/acr-hedge**), and **make gateway-deposit** (Unified Balance Kit) makes the last operator step something an agent does for itself.

THE PRODUCT

Three indices, live on Arc testnet – chain **5042002**

INDEX	MEASURES	LATEST ON-CHAIN VALUE
ACR-INF	Inference – \$/1k tokens	0.49236
ACR-GPU	GPU compute – \$/GPU-sec	0.01110
ACR-DATA	Data egress – \$/MB	0.00209

Every hourly print ships **three numbers**: the rate, a confidence interval, and the **attack-cost-per-bp** (ACR-INF $\approx$ **0.0094 USDC/bp**).

Read live from ACROracle at final submission – posted **hourly, in-cloud, signed by a Circle custody wallet** – and now the print the **futures venue freezes at expiry**.

THE HEADLINE RESULT

\$8,000 wash attack. The attacker got $\leq 2.39\%$.

INDEX	NAIVE VWAP MOVED	ACR MOVED	RESISTANCE
ACR-INF	+110.8%	+0.20%	562x
ACR-GPU	+107.1%	-1.02%	105x
ACR-DATA	+122.5%	+2.39%	51x

36,000 adversarial authorizations; the attacker burned **\$147.60** in fees. Claims are **CI-gated** (`make eval-gate`) so they cannot silently drift.

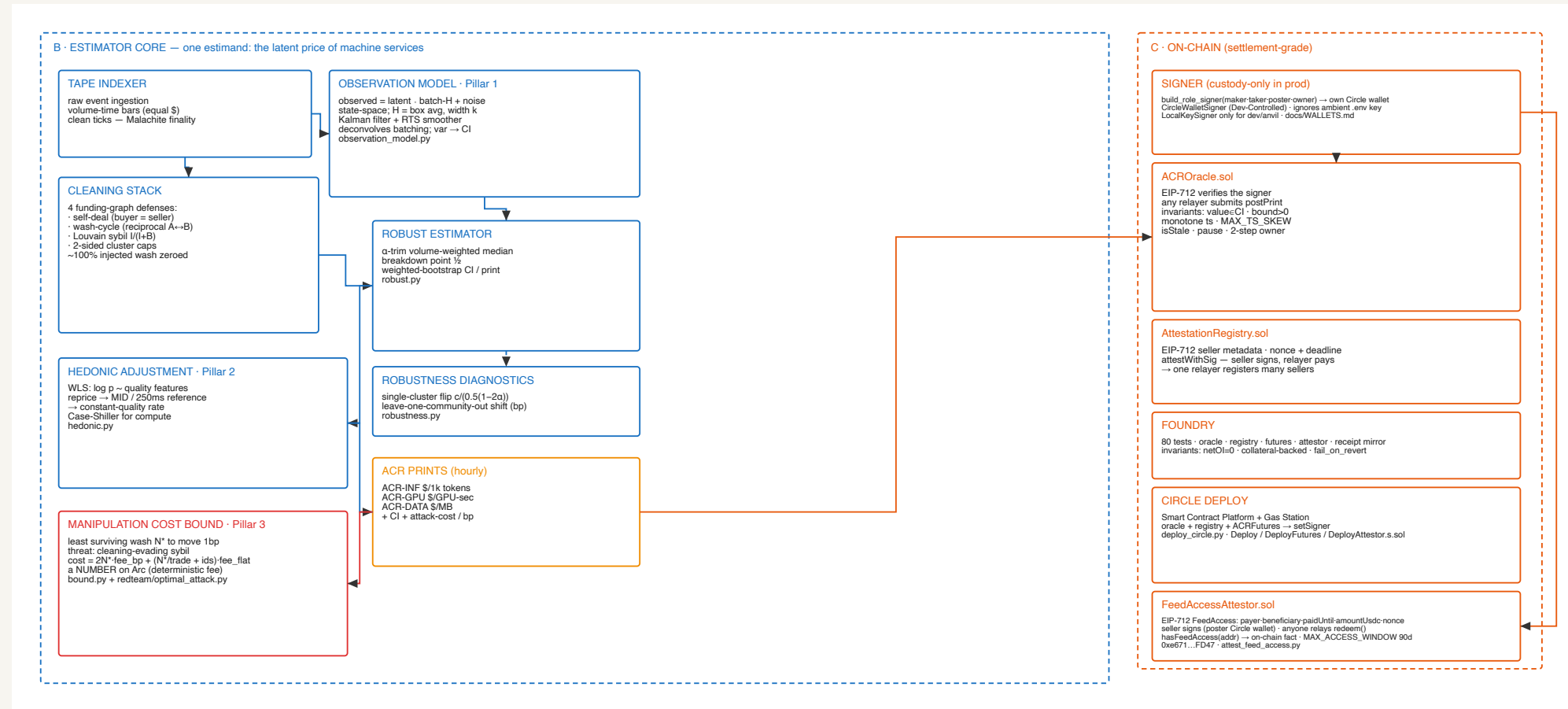
Six **mathematical** dependencies – not deployment conveniences

ARC PROPERTY	WHAT IT BUYS THE ESTIMATOR
Malachite deterministic finality	Clean tick timestamps
Gateway: one canonical rail	Complete observation, no selection bias
x402 nanopayments	Agents are the index's paying customers
USDC numeraire	No gas-token deflator in the signal
Deterministic USDC fees	The attack cost is a number, not a distribution
Agent Marketplace	Native machine-to-machine discovery

And the bridge nobody else has to build: Gateway settles nanopayments off-chain, so "this wallet paid for the feed" normally lives only in a seller's ledger. **FeedAccessAttestor** signs that receipt with the same custody wallet that signs prints, so **an off-chain payment becomes an on-chain right** – which is what lets a venue rebate fees to the wallets that paid for its index.

ARCHITECTURE · ONE PIPELINE

Adversarial exhaust in → settleable print out



Tape → clean (wash · sybil) → deconvolve (Kalman/RTS) → α -trim median + hedonic → print + CI + attack-cost → EIP-712 oracle → futures settle → agents buy the rate back via x402.

Recovering the latent constant-quality price

PILLAR	WHAT IT DOES
Observation model	Kalman + RTS smoother deconvolves Gateway's settlement batching
Cleaning	4 funding-graph defenses (self-deals, wash cycles, sybil clusters, caps) – ~100% of injected wash zeroed
Robust core + hedonic	Trimmed volume-weighted median (breakdown point $\frac{1}{2}$) + bootstrap CI; quality-adjusted via on-chain seller attestations
Manipulation bound	Prices the cheapest surviving attack per bp – validated attainable by <code>redteam/optimal_attack.py</code>

Full spec: `docs/methodology.md` – published before liquidity, the way SOFR was.

Deployed, printing, hedgeable – all seven weeks shipped

CONTRACT / SURFACE	LIVE PROOF
ACROracle 0x4f00e3BD...9092609	Hourly EIP-712 prints, Circle custody signer
AttestationRegistry 0x23ae3E1A...2f63dFb7	4 seller attestations on-chain
ACRFutures 0x29d97c62...82642fe	Three live books (INF/GPU/DATA), maker is a Circle custody wallet , cash-settles on the print
Terminal + Seller API	API x402-gated at \$0.0001/query , fail-closed

make ci green 2026-09-13: **691 py** · **156 forge** (incl. futures + invariants) · **159 terminal** · **31/31 agent** · resistance **4/4** · interop **12/12** · glossary **560/560** · GitHub CI **6/6**.

Real settlement through **Circle Gateway** – batch-UUID receipts in-repo and on the public `/exchange` tape; Circle's own CLI paid the gate (`payable`).

Provenance labeled on every value – `sim` / `gateway-ref` / `tx` . Reproduce: `make setup && make ci && make demo` .

Attack the index, then hedge on it

1. `/` – the print ticking live: rate + CI + attack cost. New to the jargon? Flip the masthead to **plain** – the whole paper re-sets in plain English.
2. `/attack` – press the button: wash flow floods in, naive VWAP swings **~+110%**, **ACR holds**, and the counter burns the attacker's USDC. Offline twin: `make demo` – one command, ~1 minute.
3. `/curve` – the term structure, the **live futures desk** read from ACRFutures on-chain, and **the Public Desk: trade it yourself**. Open a Circle user-controlled wallet in the browser (your PIN, our gas), take the stake, place a real fill, withdraw it again – the key never leaves your device, so every step is a challenge only your PIN can sign.
4. `/exchange` – **the hedger, both legs in one card, which is what this slide's title promised**.
 - prints it bought: real **Circle Gateway batch UUIDs**, \$0.0001 each, settled by the agent's own wallet – a settlement reference, not a hash, because Gateway batches off-chain. Then the joint: the venue fills every trade at that same print, so the card prints the arithmetic live – **2.47 × 0.49236 × 10 = 12.16 USDC** was the position those prints bought when this slide was cut.
 - fills it took is the on-chain half. An agent at its mandate stops trading, so when its fills have aged

Arc mainnet is September 16. This is day-one infrastructure.

- **Nothing here is a testnet exhibit.** The estimator, the oracle, the venue and the gate are chain-first; the only Arc-specific assumptions are the ones that make the manipulation bound a number at all – deterministic USDC fees, USDC as gas, sub-second finality.
- **The one thing we are missing, named:** a dependent that is not us. The venue settling on the feed is a real dependent – but we deployed both. One outside consumer turns "our two components agree" into "someone else is exposed to this number." The Marketplace listing is submitted; Circle's Discovery API carries no Arc network yet.
- **And a community, with something to hold.** Post-hackathon: **\$GPOOR** on **ACTFUN**, Arc's own community-mined launchpad – you mine it by complaining about compute prices on-chain, with the live ACR print in the complaint. It belongs after a submission, not inside one: the mining window is an hour, and a launch nobody can mine is not a fair launch.

Stop whining. Hedge.

WHY THIS WINS

The rate machine commerce settles on

- **Judge fit:** ICE – a benchmark administrator – is on Arc's testnet roster; Apollo, BNY, Mastercard are rate-native.
- **The neutrality moat:** Circle can't own the benchmark (the LIBOR lesson) – ACR is a partner, not a feature.
- **The loop closes:** resistant print → on-chain oracle → **cash-settled futures** + x402-paying machine customers → attestations → a better print.
- The first machine-commerce benchmark with a **live derivative venue** – three books, a term structure, and agents on both sides of it.

Terminal `arc-compute-rate.vercel.app` · **API** `acr-api-1fto.onrender.com`

Contracts `0x4f00e3BD...` / `0x23ae3E1A...` / `0x29d97c62...` on `testnet.arcscan.app` ·
`docs/SUBMISSION.md`